Multinational Business Finance

Continuous Assessment
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Professor Dr. Volker Seiler

USD/EUR Covered Interest Parity Analysis (2008–2025)

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1. Introduction

This project tests the Covered Interest Parity (CIP) condition for the USD/EUR exchange rate using daily data from Bloomberg for the period 1 January 2000 to 31 December 2025. Due to data availability, the effective sample starts on 3 October 2008. The analysis is performed for four investment horizons: 1 week, 1 month, 6 months, and 1year (12 months).

We chose Python for this project to challenge ourselves with data manipulation and regression analysis. The EUR is treated as the domestic currency (since we are based in Germany) and the USD as the foreign currency. Data collection from Bloomberg was initially difficult due to unfamiliarity with the terminal, but we eventually obtained the required series following the professor's guidance on using deposit rates and LIBOR equivalents.

2. Data Preparation

2.1 Data Sources

- **Spot rate** (USD/EUR) – Bloomberg ticker: USDEUR Curncy
- **Forward points** for 1W, 1M, 6M, 12M – Bloomberg: EURUSD 1W BGN, etc.
- **USD interest rates** for corresponding horizons – Bloomberg deposit rates
- **EUR interest rates** for corresponding horizons – Bloomberg deposit rates

2.2 Merging and Cleaning

The data were loaded from two Excel files: Spot_Forwards_USDEUR.xlsx (contains spot rates and forward points) and Deposit_USDEUR.xlsx (contains USD and EUR interest rates). After converting dates to datetime format and merging on Dates, we restricted the sample to observations from 2008-10-03 onward because earlier forward points were missing. This left 4499 daily observations.

2.3 Constructing Forward Rates

The outright forward rate is calculated as:

$$F = S + \frac{\text{points}}{10000}$$

We created new columns for each horizon using this formula and dropped the original points columns.

2.4 Computing Variables for CIP Regression

Following the log-approximation in equation (2) of the assignment:

$$f_{t,k} - s_t = i_{t,k}^{d} - i_{t,k}^{f}$$

log(Forward) - log(Spot) meaning EUR rate - USD rate (domestic minus foreign)

Note: The interest rates are in percent, so to obtain decimal values they must be divided by 100. In the regression we will scale the explanatory variable accordingly, i.e., use (EUR_rate - USD_rate)/100. This ensures the coefficient can be interpreted as the time-to-maturity in years (under continuous compounding).

3. Time Series Visualisation

For each horizon, we produced a dual-axis plot showing:

- Spot rate (USD/EUR)
- Forward rate (USD/EUR)
- USD interest rate (%, right axis)
- EUR interest rate (%, right axis)

3.1 Observations

1Week

The spot rate fluctuated between 0.66 and 1.04, with notable volatility during the European debt crisis (2010–2012) and the COVID-19 pandemic. The forward rate traded consistently above the spot, implying a persistent premium for the EUR. Interest rates were near zero between 2014 and 2021, then rose sharply in 2022–2023 as central banks tightened policy.

1Month

Similar patterns, but the forward premium is slightly larger. USD rates peaked at about 5.5% in 2023, while EUR rates remained negative for extended periods.

6Month

The gap between spot and forward is more pronounced. USD rates reached ~ 6% in 2023, while EUR rates became positive only in 2022. The forward premium expands when the interest rate differential widens.

12Month

Here the forward rate sometimes traded below the spot, indicating a forward discount on the USD. This suggests market expectations of USD depreciation over a one-year horizon. The interest rate differential reached historic highs in 2023 (USD > EUR by over 5 percentage points).

4. Descriptive Statistics

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================================ Descriptive Statistics for 1W ==============================
|              |      mean |       std |        min |        25% |       50% |       75% |      max |
|:-------------|---------:|---------:|----------:|----------:|---------:|---------:|---------:|
| Spot_USDEUR | 0.841265 | 0.081341 |  0.6608   |  0.7656   | 0.8572   | 0.9077   | 1.0423   |
| USDEUR1W    | 0.84113  | 0.081147 |  0.661225 |  0.765594 | 0.856818 | 0.907502 | 1.04174  |
| USD_1W      | 1.41673  | 1.76711  |  0        |  0.175    | 0.345    | 2.235    | 5.6225   |
| EUR_1W      | 0.564549 | 1.34254  | -0.65     | -0.42     | 0.07     | 0.82     | 5        |

================================ Descriptive Statistics for 1M ==============================
|              |      mean |       std |        min |        25% |       50% |       75% |      max |
|:-------------|---------:|---------:|----------:|----------:|---------:|---------:|---------:|
| Spot_USDEUR | 0.841265 | 0.081341 |  0.6608   |  0.7656   | 0.8572   | 0.9077   | 1.0423   |
| USDEUR1M    | 0.840444 | 0.080721 |  0.660894 |  0.76547  | 0.855642 | 0.906478 | 1.03992  |
| USD_1M      | 1.49997  | 1.78077  |  0.09     |  0.21     | 0.465    | 2.3      | 5.6705   |
| EUR_1M      | 0.615098 | 1.36228  | -0.65     | -0.4      | 0.08     | 1.105    | 5.25     |

================================ Descriptive Statistics for 6M ==============================
|              |      mean |       std |      min |        25% |       50% |       75% |      max |
|:-------------|---------:|---------:|--------:|----------:|---------:|---------:|---------:|
| Spot_USDEUR | 0.841265 | 0.081341 |  0.6608 |  0.7656   | 0.8572   | 0.9077   | 1.0423   |
| USDEUR6M    | 0.836213 | 0.077593 |  0.662  |  0.764551 | 0.848494 | 0.900092 | 1.02834  |
| USD_6M      | 1.84451  | 1.73918  |  0.15   |  0.47925  | 1.02     | 2.66     | 5.9575   |
| EUR_6M      | 0.839755 | 1.38178  | -0.5525 | -0.265    | 0.235    | 1.7      | 5.525    |

================================ Descriptive Statistics for 12M ==============================
|              |      mean |       std |        min |        25% |       50% |       75% |      max |
|:-------------|---------:|---------:|----------:|----------:|---------:|---------:|---------:|
| Spot_USDEUR | 0.841265 | 0.081341 |  0.6608   |  0.7656   | 0.8572   | 0.9077   | 1.0423   |
| USDEUR12M   | 0.832846 | 0.071966 |  0.677585 |  0.766759 | 0.841615 | 0.893048 | 1.01865  |
| USD_12M     | 2.05591  | 1.67009  |  0.24     |  0.785    | 1.365    | 2.87     | 6.0075   |
| EUR_12M     | 0.989769 | 1.37325  | -0.5025   | -0.15875  | 0.4675   | 2.05     | 5.51     |
```

Key Observations:

Spot and Forward Rates:

The mean spot rate (0.8413 USD/EUR) is identical across all samples, confirming we're using a consistent dataset.

As the horizon lengthens, the mean forward rate decreases slightly

(1W = 0.84111) (1M = 0.8404) (6M = 0.8362) (12M = 0.8328)

This pattern indicates that, on average over the entire sample period, the EUR traded at a forward discount relative to the USD (forward rate < spot rate), meaning the market expected the EUR to depreciate against the USD.

The standard deviation of forward rates also declines with maturity, suggesting longer-term forward rates are less volatile than short-term rates.

Interest Rate Patterns:

USD rates increase with maturity (mean 1W USD rate = 1.42% Vs 12M USD rate: 2.06%)
This upward-sloping yield curve reflects the typical term premium investors demand higher compensation for lending longer term.
EUR rates also increase with maturity, but remain substantially lower than USD rates (mean 1W EUR rate =0.56% Vs 12M EUR rate: 0.99%)

USD-EUR interest rate difference explains why the EUR generally traded at a forward discount over this period.

Volatility Patterns:

USD rates show higher volatility (sted ~1.7) than EUR rates (sted ~1.3-1.4)

The 1-week USD rate standard deviation (1.77) is the highest, reflecting the sensitivity of very short-term money markets to liquidity.

What This Tells Us About CIP

The descriptive statistics already hint at the CIP relationship we'll test formally in the next part, the persistent interest rate differential (USD > EUR) should correspond to a persistent forward discount on the EUR. The fact that the mean forward rate is below the mean spot rate confirms this directional relationship.

5. Regression Analysis

For each horizon we estimate the model:

$$\log(F) - \log(S) = \alpha + \beta\left(\frac{i_{\text{EUR}} - i_{\text{USD}}}{100}\right) + \varepsilon$$

where the interest rate differential is divided by 100 to convert from percent to decimal.

Horizon	alpha	beta	alpha_pval	beta_pval	R2	N
1W	8e-05	0.025579	0	0	0.712221	4499
1M	-0.000187	0.081821	0	0	0.94018	4499
6M	-0.001129	0.448356	0	0	0.909342	4499
12M	0.002733	1.10636	0	0	0.757721	4499

5.1 Interpretation

Alpha (α)
All intercepts are very close to zero, but their p-values are zero, indicating statistically significant deviations. For short horizons the magnitude is negligible (less than 1 basis point in log terms). For 12M, $\alpha = 0.0027$ corresponds to about 27 basis points, which may reflect transaction costs, liquidity, or persistent deviations from CIP.

Beta (β)
If CIP held perfectly with continuous compounding, β should equal the time to maturity in years:

1W: 7/365 ≈ 0.0192 → observed 0.0255 (slightly high)

1M: 1/12 ≈ 0.0833 → observed 0.0818 (very close)

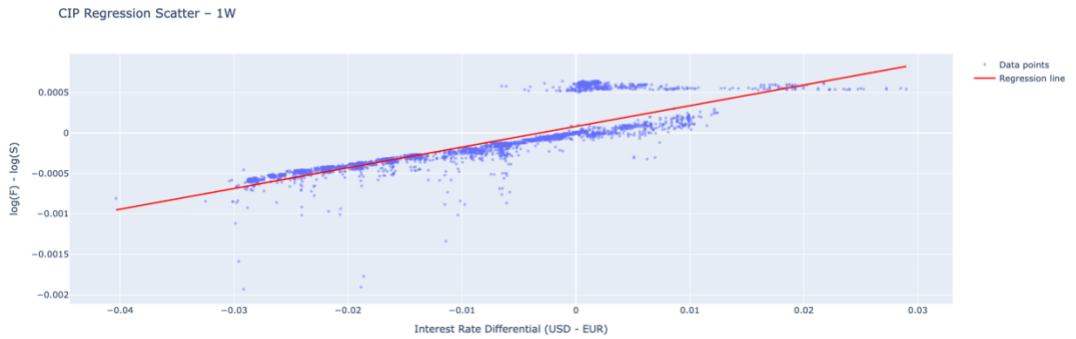
6M: 0.5 → observed 0.4483 (a bit low)

12M: 1.0 → observed 1.10636 (moderately high)

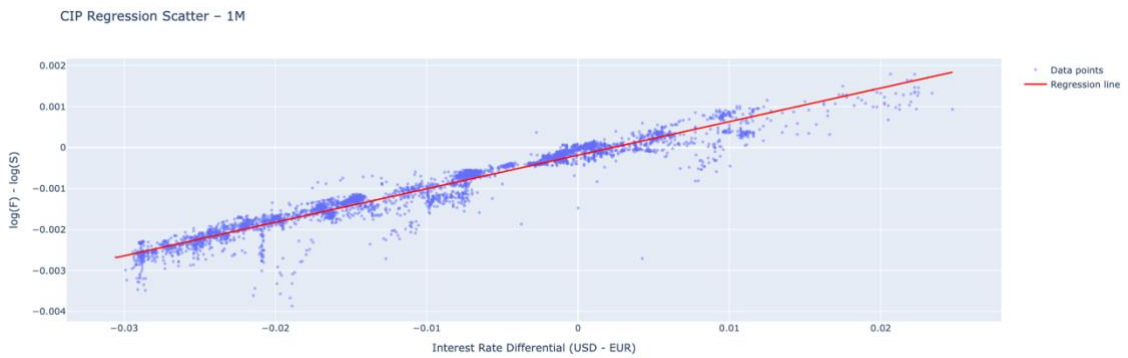
The deviations from the theoretical values suggest that the simple log linear approximation does not perfectly capture the true relationship, especially at longer horizons. The high R^2 values (all above 0.71) indicate a strong linear relation, but the slopes are not exactly the maturity.

5.2 Scatter Plots

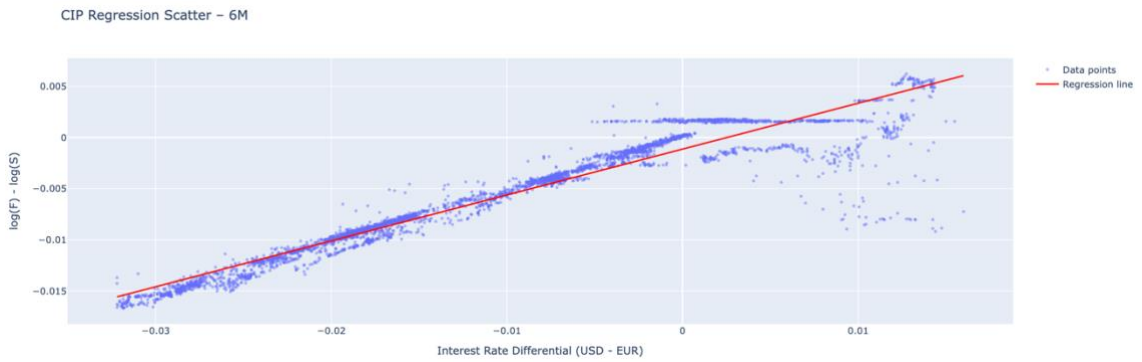
1W: Points are tightly clustered around the line, but there is a slight upward tilt. The cloud is narrow, indicating high efficiency.



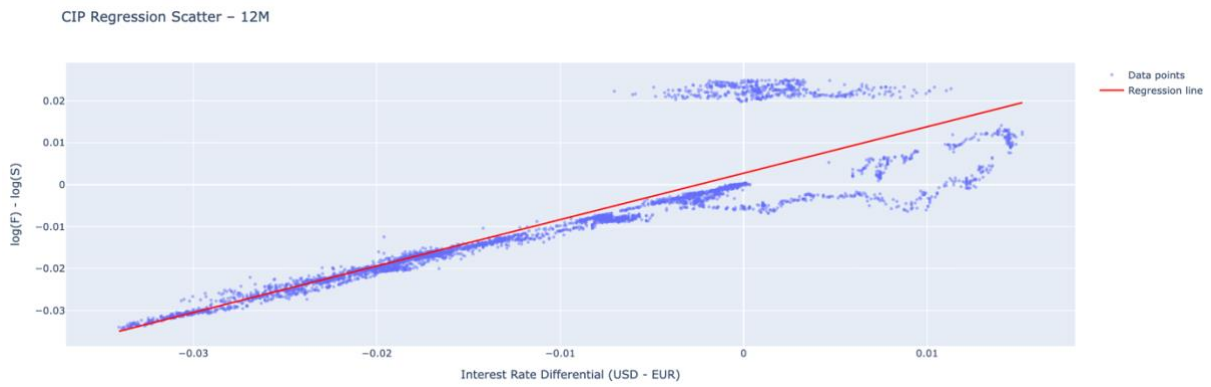
1M: The fit is extremely tight, with $R^2 = 0.94$. Deviations are minimal.



6M: The cloud is slightly wider, especially in the region of negative differentials (USD > EUR). Some points lie further from the line, corresponding to periods of market stress (e.g., 2008, 2020).



12M: The dispersion increases further. There is a clear change in slope between the negative and positive differential regions, hinting at a regime shift or a non-linear relationship.



5.3 Does CIP Hold?

The regressions show that the forward premium is linearly related to the interest rate differential, which supports the general validity of CIP. However, the coefficients are not exactly equal to the theoretical values, and the intercepts are statistically non-zero. This indicates that CIP does not hold perfectly in the data there are persistent deviations, often attributed to: transaction costs, credit risk, stress time...